

6 Institutional barriers and opportunities

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Introduction

If conceptions of how capitalism as a global system is regulated constrain change at a macro level, more local structures, cultures and competing ideas about the ordering of society also influence the development of policy, creating diversity in national and even city regulatory regimes within the broadly neo-liberal framework of ideas. How this diversity in local regimes is generated and maintained in the face of the universalising processes of neo-liberalism can be explained in a number of ways: local path dependencies, professional cultures, definitions of problems, opportunities and risks.

The continuity of public policy from one political administration to the next is something most people take for granted. Root and branch change of policy with each change of government would be chaotic if it were not also impossible. So continuity need not be seen as necessarily a policy failure. Rather this continuity turns into policy failure when the path of policy development becomes inconsistent with the long-term public interest, or when that which is holding matters stable becomes a barrier to publicly desired change. The task of change suggested in Chapter 4 therefore involves a path transition, which not only generates new policies, but challenges those structures which prevent adaptation to new external circumstances – for example, climate change. This task requires an understanding of what holds us to our current path, recognizing opportunities and turning points when they present themselves, and redirecting us to new destinations.

This chapter commences with an analysis of the evidence for path dependence in transport policy using case studies from Australian cities (Low and Astle, 2009; Curtis and Low, 2012). The second section is concerned with forms of action and thought that are constrained by the ‘art of government’ in play in particular situations. The limitations imposed by specific arts of government are considered through an investigation of mega urban transport projects (Dean, 1999). The third section concerns limitations on the governmental and community capacity to select new pathways imposed by conceptions of risk. Finally we consider governmental capacity to implement policy in terms of negotiations with the public and between agencies and levels.

Working with these concepts means foregrounding the ideas, processes, rationalities, technologies and organizations that are already present and which act to maintain the current path. Rationalities are generated which constantly reinforce what is already in place. The path to a new sustainable transport system leads not through ‘green field’ sites of public policy, but rather through policy ‘brown fields’ which are already densely populated by human mentalities. Change will occur only after sufficient dismantling of what is already present to make room for the new. As was clear from the previous chapter, and as we shall see in Chapter 8, such transformation occurs through the agency of persons and groups.

Path dependence

Research by Curtis and Low (2012) on barriers to sustainable transport uses the ‘path dependence’ concept as a theoretical framework to explain why transport policy in Australia endures beyond its rational basis and how it could be responsive to change in a geophysical context. Path dependence began in the field of economics to describe aspects of the market which did not seem to be driven by the rational logic in which supply is driven by demand (Arthur, 1994; Pierson, 2000). It was found rather that, for some products, chance followed by the incentives of real production systems created the momentum to keep producing products even when better ones became available. There are two main claims involved.

- 1 The presence of a product on the market has a history and can only be understood in a historical context. Circumstances or a confluence of events at the time of initiation of production determine the path taken.
- 2 Sometimes there is more value to a producer to keep producing essentially the same product – albeit continually improved – rather than change to a different and superior product.

These ideas have been applied to institutional systems that generate public policy by Nobel laureate Douglass North (1990) and political scientists Paul Pierson (2000, 2004) and Jakob Torfing (2001). Consequently in applying the concept of path dependence to transport systems in Australian cities, the research undertook to build an understanding of the historical context in which transport systems arose, and to combine that with an examination of what advantages there may be in maintaining a particular type of system rather than changing it.

Under the theory, path dependence occurs where there are:

- large setup or fixed costs
- learning effects which lead either to better ways to produce the product – reducing costs – and/or consumers learning new and better ways to use the product

- coordination effects so that the benefits to a singular person using the product increase as others adopt it
- adaptive expectations created when it is widely understood that the product will continue to be available and therefore more people take it up.

The research found, through a study of three State capital cities – Sydney, Melbourne and Perth – that these elements are present in Australian transport systems that perpetuate a largely private system of transport based on the motorcar. Clearly there are large setup and fixed costs – notably infrastructure – in the transport system, both for mass transit systems and motor vehicles. There are learning effects: for example the development of private sector engineering firms' expertise in road transport infrastructure design and travel behaviour projection. Learning extends to marketing where the private motorcar has become a symbol of freedom, and to the use of vehicles when transport consumers find new ways to use their vehicles, from travelling further distances through to the development of park and ride schemes. Clearly in the massive public investment involved in the transport system, each individual's capacity to use their car is improved when others take up that mode, because it justifies increasing road building to meet demand evidenced by traffic congestion. So far, governments in Australia have sought to meet that demand rather than allow the experience of the driver to deteriorate as more drivers push road capacities to the point of congestion. Meeting that demand has required massive public expenditure on motorized transport in the form of ever larger and more complex road networks. This, in turn, creates the expectation that such investment will continue and that the motorcar will remain the transport mode of choice.

Thus the current transport system can be characterized as path dependent. This characterization does not, however, reveal much about how this situation came about. Nor does it illuminate what is required to break open the institutional carapace of organizations' policies, beliefs and practices that has been built around the vision of free mobility via the private motorcar. Transport systems and their use result from considerable physical, social, cultural, financial and ideological investments – investments that also serve to protect such systems from external challenges.

Curtis and Low (2012) conclude that the technical dependence of cities on car transport has been shaped by continuous investment in bigger and better roads, which encourages land use patterns serviced by the car. However, the specific circumstances which led to this technical manifestation are complex, and have, among other things, been shaped by a strong, unified, institutional structure for road planning and construction coupled with a weak and fragmented structure for public transport (see also Low and Astle, 2009). This strong and unified organizational structure has been underpinned by a remarkably consistent discourse on the benefits of private motorized transit, a discourse of freedom and modernization. They found

that path dependence in the transport system in Australia is thus maintained through three elements:

- the momentum of physical infrastructure
- institutional power determined by well-integrated agencies with the capacity to develop far-reaching policy
- discourse that defines – and thereby limits – the conception of the problem. Policy is shaped by arguments and perspectives, or storylines, supportive of the current direction of policy.

The momentum of physical infrastructure operates in two ways. First, transport choices are conditioned by opportunities, which are embedded in the infrastructure provided. Thus people located in suburbs not serviced or poorly serviced by public transport, but with significant road infrastructure, will travel by car. The assumption of transport modelling is that demand is independent of supply. Yet the now widely accepted postulate of induced traffic implies that supply – of road space – itself generates demand (SACTRA, 1994, 1999; Goodwin, 1996).

Second, the constant production and improvement of the road network perpetuates consumers' belief that the car is here to stay, and thus is the mode of choice for long-term transport needs. Transport and land use become inextricably linked when a high speed road-based network is built, thus enabling urban structuring catering for large car-based catchments with increasing distance between home and work and urban services. This tendency to fractured and dispersed urban form diminishes the possibility of access by other slower modes (see Box 6.1).

Institutions are the framework of rules and norms for policy making. Institutions both limit intentional action and make action possible by providing definitions of problems, solutions to those problems, and the knowledge to implement those solutions. Through organizations they create a corps of personnel bearing that knowledge who can implement solutions. The research has found that in Australian cities, the relative power of transport agencies is unbalanced (Low and Astle, 2009: 50–1). Road agencies have a high degree of control over the full gamut of activities and functions relevant to design, construction and maintenance of the road network. They have direct lines of reporting to both State and federal ministers, and high levels of expenditure and discretionary control over that expenditure. In contrast, public transport agencies have traditionally been disenfranchised, with policy, design, construction and operations split between different agencies (some privatized), little direct access to ministers, and a debt burden that cripples their capacity to expand operations without explicit ministerial approval.

Finally, there are significant storylines or supporting discursive propositions that create a shared language for support of road investment in Australia. The research found that there is a high level of agreement about these storylines; even where the conclusions about what should be done

Box 6.1 Technical path dependence

The vast investment in the physical structure of cities itself casts its shadow over the future (Arthur, 1988). Nobel laureate Paul Krugman (1991: 80) observes:

If there is one area of economics in which path dependence is unmistakable, it is in economic geography – the location of production in space. The long shadow cast by history over location is apparent at all scales, from the smallest to the largest – from the cluster of costume jewellery firms in Providence [USA] to the concentration of 60 million people in the North East Corridor.

The physical form of a city can be shaped by the dependence on a particular form of transport (Schaeffer and Sclar, 1975; Newman and Kenworthy, 1989). In the case of Australian and most North American cities, and the suburban extensions of European cities, the use of the private vehicle for getting around and servicing the city has made possible the choice of low-density residential environments. In turn these low-density environments encourage the building of high-quality roads to accommodate private vehicles, thus reconciling low density with reasonable accessibility to services. (Cervero, 1998: 29) observes for American cities, ‘The scattering of activities to all corners of the modern metropolis poses unprecedented challenges to public transportation’. In later work he argues that road investment and urban development are co-dependent (Cervero, 2005: 128–9).

We would argue, however, that such technical path dependence – in the form of urban density – is itself produced by institutional path dependence, and is not a completely determining factor (as also argued by Mees, 2010: 53–5). Once institutional path dependence is overcome, a gradual transformation of transport systems can occur which can be made to serve whatever city form has been laid down over time. We cannot do much to change the shape of cities but much can be done quickly to change the institutions governing the creation, management and delivery of transport systems.

in the face of them is challenged. The core road-based transport storyline is one of ‘freedom and independent mobility’. The core public transport storyline is one of ‘containing the public transport deficit’. (Curtis and Low, 2012: Chapters 8 and 9).

Transport policy in Australia then is a product of the historical growth of the system, and the different capacities of the organizations responsible for developing, building and maintaining it. It is the particular competencies of policy actors that lead to path dependence, not a failure of competence. Accordingly if transport systems in Australian cities are to be taken in a new direction then there is a need to reframe the problem and redirect the competencies. However the research discussed in the next sections of this chapter indicates that this process is far from straightforward.

Art of government

Sturup's investigation into the art of government of mega urban transport projects (MUTPs) was undertaken based on detailed review of three Australian urban mega projects, the Melbourne City Link motorways, the Perth to Mandurah railway and Sydney's Cross City [road] Tunnel (Sturup, 2010 and see Box 6.2).

Mega projects consistently run over time and over budget (Flyvbjerg et al., 2003). This outcome endures despite comprehensive work and

Box 6.2 Case studies of Australian transport mega projects

Melbourne

Melbourne City Link is a 22-kilometre toll road connecting three free-ways around the central business district of Melbourne. The road includes upgraded sections of already existing freeway, an elevated section of roadway, a bridge and two tunnels. City Link was built under a Concession Deed granted to a consortium of Transfield/Obayashi to design, build, finance, operate, levy tolls and maintain it for 34 years until 14 June 2034. Ownership will then transfer to the Government of Victoria (Infrastructure Partnerships Australia, 2006). The project cost AUD2.6 billion.

Perth

The Perth to Mandurah railway – also known as the Southern Suburbs Railway or South West Metropolitan Railway – is owned and operated by the State of Western Australia. It stretches 70.1km from central Perth to Mandurah in the south. The line includes two major river crossings, two sets of tunnels, construction of two underground stations as well as numerous above ground stations. Much of the line passes along the central median of the southbound freeway. The operation of the railway is undertaken by Transperth a part of the Western Australian Government's Public Transport Authority. Construction commenced in January 2004 and the line opened on 23 December 2007. The cost of the project is estimated at AUD1.184 billion (Longhurst, 2008).

Sydney

The Sydney Cross City Tunnel project was built under a public–private partnership between the Government of NSW and Cross City Motorway Company. The contract included three stages: design, finance and build twin tunnels – 2.1 kilometres in length following roughly the line of William St; design, finance and build changes to the surface streets – to ensure the decongested streets did not induce further traffic; maintain and operate surface street changes and the tunnel for a concession period of 30 years and 2 months (Catalyst Communications, 2003).

improvement to the key mega project technologies of stakeholder consultation, benefit-cost analysis, engineering, private-public sector contracting, project governance, and project management. Consideration of the art of government of MUTPs was thus an attempt to gain new insight into the underlying conditions that generate and reproduce this outcome.

Following Foucault's theory of governmentality (Foucault, 1991), the art of government of MUTPs can be seen as the amalgam of technological potentials – including the actual products of engineering – the knowledge used in producing them and the rationalities used to justify them, which together provide the ontological reality – or way of being – of the agents within the project, and thus what is possible for them to contemplate doing.¹ Another way of talking about this ontology is that it is the occurrence of the world. Humans act in harmony with the way the world occurs for them. Thus, in MUTPs, actors conduct themselves in harmony with the way the art of government of MUTPs allows the world to occur to them.

This research suggests that the intractability of cost and time overrun in MUTPs is not a product of having the wrong solutions – for example, better costing of externalities in benefit-cost analysis – nor is it a product of the solutions being underdeveloped. Rather, it is suggested that the underlying art of government in which the solutions to these and other problems are developed is different from that in which MUTPs occur. This means that when the solution is used in the MUTP it is adjusted to conform to the MUTP world view. The solution and the MUTP therefore do not 'speak' to each other very well. Within the MUTP, the solutions do not occur as correct, or appropriate, and they are therefore adjusted to support the world view of the MUTP. Sometimes this disjuncture results in solutions seeming oddly incompatible, as though they have been bolted on from the outside as, for example, stakeholder engagement often occurs in mega projects.

Sturup's (2010) analysis found four discernible features of the art of government of MUTPs. The art of government of MUTPs is built from the art of government of 'project' in a more general sense. The art of government of MUTPs is technical. It is pharaonic – or sovereign. And it is organically sticky.

The art of government of project stems from the definition of a project as a unique activity, to be produced within a specific timeframe. In operation it works to ensure that that definition is validated. The technologies of project are designed to maintain the uniqueness of the project and to focus attention on delivery within a timeframe. To achieve this result, the project must be defined within set boundaries of what *is* in the project and what *is not* in the project. Art of government of MUTPs shares these features with art of government of project. MUTPs must therefore be defined as unique activities bounded by time which are also colossal, complex, controversial, captivating, costly, and have control issues (Trapenberg Frick, 2008). An MUTP adds to this definition the construction of some element of the transport system. Like the art of

government of project, the art of government of MUTP stems from this expanded definition and similarly operates to ensure that this expanded definition is validated. The control issues referred to by Trapenberg Frick (2008) are a result of needing to place bounds around the activity in order for it to be deliverable within a timeframe. Because of their size and the complexity of their impact, this process of bounding the project is much more complicated for MUTPs than for small projects.

The art of government of MUTPs is in essence technical. Because an MUTP by definition involves the construction of a road, railway, bridge or tunnel, the project is necessarily a physical engineering *solution*. MUTPs are thus technical solutions in search of a problem. If the problem is to be solved by the construction of some physical element of the transport system, essentially by an engineering solution, then the problem needs to be articulated as solvable by that solution. The process of constructing a problem together with its solution is therefore also a process of rendering complex social issues into technical engineering problems that are subject to control. This reification is central to the art of government of MUTPs. Many issues internal to the project are similarly rendered technical issues, such as the allocation of power to act in the project, setting the limits of stakeholder engagement, and ensuring assessment of the project is restricted to whether the project achieved its limited goals such as, for example, delivering the defined object on time.

The art of government of MUTPs always involves sovereign power over decisions. The research showed that all the MUTPs investigated were considered to start at the point at which someone with the capacity to deliver it, promises to do so assuming that the parameters can be satisfactorily worked out. Thus, in projects, power is circumscribed by the technology of project. Power to act in the project becomes a function of the decision to move forward, and is moderated according to the project planning documentation. Individuals in the project are given the power to act by virtue of the decision to go ahead with the project. Thus the allocation of power, and its function in the project, is rendered apolitical and technical. For this process to work, the originating decision must be made by someone who is not only capable of delivering the project, but who is also in a position to decide convincingly that the project should happen. Such an actor also needs to be able to defend this position. The logic of power in the art of government of mega urban transport projects is therefore pharaonic or sovereign – ‘I am the king, my will shall prevail’.

Many of the features of MUTPs are designed as a result of, and to guard, this logic. An example is the technology of stakeholder consultation. In MUTPs it is impossible to allow authentic consultation to occur, because such consultation must allow for the possibility that the project should not happen. To question whether the project should occur would unravel both the construction of power within the project and bring into question the power of the decision maker to make sovereign decisions. Since neither outcome is acceptable, the use of authentic stakeholder consultation to

alleviate difficulties relating to public protest, which in turn lead to delays and cost overruns, cannot be effectively implemented.

The art of government of MUTPs is organically sticky. In tracking the development of the rationality of the problem and the rationality of the solution of the three projects, it becomes apparent that not all the rationalities deployed in the construction of the problem and its solution are created after the sovereign decision which started the project was made. Indeed the construction of the problem and solution may have started some time before, and makes use of established rationalities in the context in which the project is to be built. For example in Perth, the rationality used to justify the Mandurah project was that which was used to justify the railway extension – the northern suburbs rail line – established after the reopening of the Fremantle line some 20 years earlier. In this sense MUTPs are *sticky* in that rationalities stick to them.

While the rationality of the problem and solution are evolving together, the rationalities used to justify changes to the project, and hence to the understanding of the problem, and why it needs solving, tend to hang around and re-emerge, almost never being entirely abandoned. This finding is not dissimilar to that of Miller and Lessard (2000), who also found that projects evolve. It suggests that like Machiavelli's prince (Machiavelli, 1979 [1532]), the sovereign may not make unilateral decisions simply on a whim. Decisions must be justified in terms of the greater good. In the art of government of MUTPs these sticky rationalities bleed around the decision point, so that absolute demarcation of the project within a particular scope and timeframe is not possible. The result of this stickiness is that even where the decision makes no explicit reference to a particular rationality, that rationality nevertheless sometimes remains a driving factor in the project.

Sturup's work on the art of government of MUTPs provides an alternative mode of analysis to that of path dependence in explaining the stasis of local policy positions. It provides a way of considering the historical development of institutional structure and policy in terms of what world view the rationalities, knowledge and technologies that are used in the project continue to generate in the present. Such world views are the source of being, and define what is logical or 'right' action. Any attempt to adjust the structures that maintain path dependence must be done with consideration for the mode of *being* – ontology – that the new structure will generate. Adjustment will only be successful if the new amalgam of rationality, knowledge and technology is distinguished in such a way that it can be understood as a step change from the world view generated in the past.

Conceptions of risk

Within an overall study about the influence of the conception of risk on children's independent mobility, Rudner (2011) has tapped into another level at which to consider why the transport systems of Australian cities resist change. Together, policy professionals and the lay public create a public

'knowing' of risk that promotes risk aversion rather than exploration and experimentation. Each policy system has a different risk rationality based on the focus of the organization, the problems identified, the methods of data collection employed, and the analyses, interpretations and solutions applied to the issues it faces. Meanwhile there are also differences in risk rationality among experts, and between experts and the lay public.

When we consider transport systems, risk management practices are used to reinforce and maintain current systems, and therefore path dependence. An example is the approach to 'road safety' described in Chapter 4 in which the risks of automobility in the USA were shifted from the transport system to individuals and their behaviour (Packer, 2008: 28–31). Decisions about how to deliver policy outcomes are strongly influenced by technical approaches to risk. Technical approaches are applied from planning and construction to public education on road safety. For example, proven competencies in road building are more likely to pursue low risk projects in road construction than expand into unknown territory such as building new railways. Crash statistics that demonstrate that children are more likely to be involved in road trauma than adults are used to justify control over children rather than control over traffic speeds.

Within many policy and professional domains, risk is considered an externally imposed hazard that impinges on a person or organization; these hazards and their potential consequences are believed to be best evaluated by experts. However, risk assessment experts are trained and socialized within the systems in which they operate, limiting the capacity for organizations to change their approach to risk assessment of projects. For example, Australian road safety auditors consider safety primarily from the 'windscreen' view rather than from the view of the pedestrian or cyclist. To move beyond narrow technical understandings of risk, Rudner suggests an alternative way of conceptualising risk that releases the individual's capacity to look after him or herself. Founded upon an ecological psychology perspective (Gibson, 1986) she provides a pragmatic conception of risk as an interactive relationship between humans and their environment that incorporates notions of agency and competence.

This re-conceptualization asserts that risk is a relationship between the intent of an agent (individual, group, or organisation), what the environment offers [in the form of opportunities and dangers], and the capabilities of an agent to use the environment in the intended manner. ... Trade-offs between possible costs and benefits in order to potentially achieve desired outcomes are part of ongoing decision-making processes, or stream of action, in the agent–environment relationship.

Rudner, 2011: 20

This new conception of risk is underpinned by the notion of 'affordances'. Affordances comprise the ambient array of objects in the environment that offer action possibilities, and individuals' abilities to perceive and actualize

these action possibilities. When individuals actualize perceived affordances, further action possibilities within the environment are revealed, expanding perceptions of affordances and the capabilities to use them (see Box 6.3). However, individuals' use of environmental affordances may be constrained by social and cultural expectations. Through socialization we learn 'what affordances can be utilized, *by whom*, and *when*' (Reed, 1993: 52, emphasis in original). Our actions therefore, are circumscribed by individual 'specializations' such as age, gender, and social status, all of which influence our behaviour, directly and indirectly, although social and cultural expectations may be challenged.

In her study of children's independent mobility, Rudner (2011) found a variety of policies in which assumed hazards and associated risks are assessed as the same for all children. No account is taken of variation among children's different individual capabilities and experiences in relation to the situations they may encounter. Static understandings of risk perpetuate a causal approach – from hazard to victim to damage – that assumes incompetence on the part of children as well as adults and professionals. This associated incompetence arises through a policy focus on precautionary principles, worst possible outcomes, and regulation that defines adult and professional responsibilities and formalizes the education required to comply with policy objectives concerning risk.

Conceptions of risk and how it is managed are a reflection of broader social and cultural norms, expectations and social control, as well as localized differences. Fyhri et al. (2011) found differences in parents' assessments of the dangers children might encounter if they go places by themselves, as

Box 6.3 An example of different affordances arising from different competencies

The 'affordance' includes the particular element of the environment – for example, a staircase – together with the competencies of the person in that environment. For instance, while non-skateboarding, able-bodied individuals may only perceive the affordance of walking up or down stairs, the capabilities and previously actualized affordances of a skilled skateboarder allows them to enlarge the possibilities beyond walking the stairs, to using the skateboard for jumping the stairs, or to slide down the railing.

When affordances are conceived through risk assessment processes, it is possible to say that the trade-off between the possibilities of a successful and an unsuccessful slide down the railing on a skateboard is different for different people. The trade-off is made by the person before doing the sliding, and at that point is only possible or potential. There will be a high risk of a non-skateboarding individual falling off the railing if they try to slide along it. The risk of falling is lower for someone who has rudimentary skills in this trick. For a highly experienced skateboarder, the risk of falling off is much lower than those skateboarders with less experience, or non-skateboarders.

did Rudner and Malone, and Rudner (2011) in their review of research from Australia, South Africa, Japan and Tanzania. For example, more parents from Japan indicated children were less likely to encounter potential hazards if they go places by themselves than in Australia or richer parts of South Africa, and children's independent mobility was more similar to the independent mobility of children in Tanzania (Behrens and Muchaka, 2011; Bwire, 2011; Driandra and Kinoshita, 2011). Essentially conceptions of 'risk' are not universal. Risk to children varies markedly among different countries and at different times. For instance Malone (2012: 13) describes her experience in Tokyo where she encountered groups of kindergarten children walking home from school and crossing busy roads. When she expressed her concern to a Japanese colleague that there was no adult supervision, the colleague responded, 'What do you mean, no adults? There were the car drivers, the shopkeepers, the other pedestrians. The city is full of adults who are taking care of them'. This is a very different attitude to risk and affordance than exists in countries such as the United States and Australia. Where other people are considered to be helpers in Tokyo, elsewhere they may be considered neutral at best or a threat at worst.

In Australia, a single rationality tends to be promoted: risk aversion. Other rationalities that value the costs and benefits of particular activities differently from official policy, or do not view a potential hazard or situation as a risk, are regarded as ignorance that can be rectified with statistical information, rational logic and education. In the case of children's independent mobility, the need for adults to protect children from potential traffic accidents is perceived as appropriate. Whereas children's development, such as the need for exposure to traffic in order to learn self-protective skills, or the need for children to gain a degree of independence, is not considered within risk assessment calculations (see also Chapter 10 of this volume).

While different risk rationalities are viewed as a challenge of behaviour change and management for children's independent mobility, in other areas of transport policy, different rationalities or ways of doing things are viewed as a cost rather than as a behavioural issue. However, both situations are similar with regard to the matter of professional discretion. In matters relating both to children's independent mobility and to transport projects, professionals are viewed as competent insofar as they abide by existing structures and regulations rather than utilize their discretion.

Organizations considering different pathways in transport policy have a tendency to overprice risk in projects that are seen as outside the capacity of the organization concerned, but such projects may nevertheless be within professional capacity. This occurs even though the risks can be greatly reduced by using staff who have appropriate competencies. The result in transport terms is that generally road projects carry a significantly smaller risk premium than rail projects.²

Rudner's work points to new ways of understanding the effect of continued investment in road infrastructure, and the degree to which adaptive

expectations form our interaction with the world, our conception of risk, and of correct behaviour. Affordances may offer a variety of action possibilities, but how humans relate to action possibilities is enmeshed with social and cultural structures. Both Costall (1995) and Heft (2001) emphasize that knowledge is embedded within the material environment through the discovery and use of tools, the design of artefacts, and the social representations of the material world. The built environment contains the intent of its designers, and the social and cultural intentions of the society. Although individuals may perceive and actualize a number of affordances in the physical environment based on their own objectives and capabilities, they are also guided by the knowledge contained within the material and symbolic landscape. This creates an enlarged public knowing over time and through space between the environment and individuals, a point we explore in the next section. Not only does a continued investment in motorized transport provide increased material opportunities for the use of the car; it also reinforces a cultural commitment to the continued use of the motorcar.

Connecting the public with the planners

One form of resistance to change comes from the broad public. Resistance can result when the policy that the government is trying to implement does not feel ‘owned’ by those on whom it will impinge – namely the public. While policy ownership is essential to the successful implementation of policy, engaging the community also enables tacit knowledge and people’s experience of their city to shape the way we understand the city and the substantive planning challenges it faces.³ One tool that assists in shaping many cities is the metropolitan strategic plan. Since gaining prominence in Europe – with the 1944 Plan for Greater London among the first – metropolitan plans have become widely regarded for their socio-spatial representation of land use and transport infrastructure, their potential to be either static or dynamic, and their ability to combine social and ecologically based objectives to guide decision making (Albrechts, 2006: 1153; Healey, 1997: 13).

In Australian cities, as is the case in most US cities, such plans guide decision making but often exist without strong legislative and regulatory support. In a policy context where metropolitan plans do not carry significant legislative weight, their influence results from the level of legitimacy achieved through an inclusive and deliberative process of plan making. In seeking to understand how metropolitan strategic plans might achieve greater certainty with respect to implementation, Legacy’s research (2010a) drew upon the deliberative democracy literature and on Arendt’s (1978) and Benhabib’s (1988; 1992) conceptions of enlarged thought to explore the nuanced relationship between process and legitimacy. Enlarged thought means the opening up of different perspectives through a process of open engagement (Benhabib, 1992; see Box 6.4). Inclusiveness and deliberation are highly regarded for their capacity to reduce the incidence of conventionally

constrained and predetermined rationalities in determining policy. Drawing on the deliberative democracy literature, an inclusive process promotes ongoing interaction and minimizes power imbalances among stakeholder actors (Habermas, 1990) and can be described as a procedural approach to addressing the range of challenges, including the barriers to implementation. Legacy's study looked at the process of producing enlarged thought in planning the urban environment, but in order to understand the impact that enlarged thought could have on policy decision making, the way in which these processes interface with the more formal institutional planning arrangements was examined. The result was an understanding of the connecting points that bring these two processes together and in particular the unique role that the professional planner plays.

Using metropolitan strategic plan-making processes as the focus of the study, Legacy examined these processes in Greater Perth and Greater Vancouver and paid particular attention to the way the general public, professional planners and politicians were brought together with the aim of negotiating an outcome in the context of the broader planning institutions. The relationship between the planning process and legitimacy was central to this research and was inspired by the need to address the many contemporary challenges confronting cities so that their ecological sustainability is not further compromised.

By engaging with the deliberative democracy literature, the research articulated the importance to the outcome of different ways of knowing. In the process of plan-making, different kinds of knowledge – the community's tacit knowledge and the expert's professional knowledge – are generated

Box 6.4 Enlarged thought

Enlarged thought is a concept arising from the work of Hannah Arendt (1978) and later explored by Seyla Benhabib (1988, 1992). They argue that 'enlarged thought' is the knowledge that emerges from processes that bring people together to reflect on and to gain greater understanding of the many different perspectives that exist on issues. Inclusive dialogue among a range of people, each possessing different sets of knowledge, fundamentally supports the production of enlarged thought: 'the more human perspectives we can bring to bear upon our understanding of a situation, all the more likely are we to recognize its moral relevance or salience' (Benhabib, 1988: 43).

Enlarged thought translates into the opening of different perspectives through some 'dialogic' or 'discursive' ethic (Benhabib, 1992). This 'opening up' is in direct contrast to the kinds of inward reflection and bounded thinking that occurs when the process is restricted to only a few. Enlarged thought requires that the voice of others be expressed (Benhabib, 1992) and is supported through an inclusive process that enables a broader understanding of the challenges, and reduces the impact of instrumental rationality in determining policy.

and exchanged (see Lindblom and Cohen, 1979). By mapping the knowledge exchange over the course of the metropolitan plan-making process it became clear that the knowledge generated is fragmented. Naturally metropolitan plan making is characterized by a lengthy engagement process, and the time needs to be taken to give a range of different actor groups from across the metropolitan region the opportunity to engage with each other at each stage in the process (Legacy, 2010a, b). To retain the deliberative component of the process, each engagement event is usually quite small, and in many cases focuses on the engagement of a single stakeholder group. The research revealed that while the production of knowledge was fragmented, governance arrangements are the ligaments uniting deliberative events with each other, and connecting the broader processes of stakeholder engagement with the formal structures of metropolitan strategic plan making.

Governance is critical to building the legitimacy of the plan as it is the connective tissue between the individual stakeholder engagement events and the broader institution of planning. The critical actor in any governance arrangement supporting a plan-making process is the planner. Not only do planners bring their expert knowledge to the processes, but they too possess tacit knowledge, that is their knowledge of the various barriers to plan implementation that need to be negotiated during the plan-making stage rather than at the implementation stage (Legacy 2010a, b; 2012). A legitimate plan is one that everyone owns (Irazabal, 2005: 52), but it is also a plan that can be implemented.

Legacy's research revealed that substantive impact upon policy, and indeed long-term implementation of policy, is determined not only by how people are engaged, but by the influence that different kinds of knowledge have on policy. The relationship between the knowledge of the public and the knowledge of the professional planner can be reconciled through engagement events that bring these parties together. In the process of developing and implementing a new pathway to sustainable transport, a process will be required to nurture this relationship, and gain the benefits of building legitimacy into the plan.

Government capacity to integrate policy across departments and between levels

Path dependent systems display a large degree of internal coherence. The organizational and discursive institutions that support a particular outcome are highly correlated. This is reflected in the degree of integration of policy – whether explicitly or implied – horizontally across agencies within each level of government, and vertically between levels of government. A constraint on the implementation of a new pathway is, therefore, partly dependent on the ability to create vertical and horizontal integration of new policies sufficient to produce a new transport system. An indication of this capacity can be illustrated by published government policy, both in documents and in the processes through which written policy is enacted.

Policy as expressed in published policy documents, ranging from statutory planning schemes to broad strategic visions, rationalizes planning and decision-making processes within government. Policy in this sense contributes to the framework of norms that guide processes and practices. However, published policy is often a statement of intent or aspiration, and so does not by itself provide an indication of whether governments have the capacity to plan new pathways for transport. The real capacity of policy to address sustainable transport objectives can be restricted by various barriers as published policy is implemented.

In a study of the capacity of government to integrate land use and transport planning in Perth, Western Australia, and Melbourne, Victoria, Curtis, Armstrong and Babb (2010) looked at both these aspects of policy, integration and implementation. An analysis of policy capacity as expressed in published land use and transport documents in Perth and Melbourne was the first stage of the research. In Australia, land use and transport planning is primarily the responsibility of two levels of government – State and local – which have differing powers and geographic scales of interest. Planning undertaken by the State governments at the regional level is an important means to coordinate development, deliver infrastructure and integrate transport networks. State legislation requires local government to create planning schemes that guide the use and development of land within their jurisdiction. Policy capacity in this research was seen as that of integration of policy messages across two dimensions. Vertical integration occurs across differing levels of government while horizontal integration involves policy interactions across sectors within the same level of government. Ideally, land use and transport policy should be well integrated across both the vertical and horizontal dimensions in order to provide the capacity to effect change towards more sustainable modes of transport.

A content analysis of relevant policy documents at the State and local government levels in Perth and Melbourne was undertaken, evaluating policy statements against a comprehensive list of planning principles for land use and transport integration (derived from previous research, Curtis, 2005). Although there was good overall coverage of the principles, integration across State and local government policies was uneven. Vertical integration of policy – between State and local levels – was inconsistent, with gaps in areas such as support for frequent and reliable public transport services and the management of parking. Some criteria that supported effective traffic management were supported in State level documents but had conflicting policy objectives at the local government level. On the other hand there were categories of criteria that were well integrated across the State and local levels, especially in providing for higher density development in close proximity to complementary activities: mixed land use; and the integration of scale and character within precincts. The strength of policy statements at the State level was weaker overall than at the local government level, which, in several cases demonstrated high levels of commitment to the principles. This raises the question of whether the capacity

for innovation and responsiveness to local conditions may be compromised when policy messages from the State level, which are prioritized both in their placement in the format of the document and by statute, are weaker than those at the local level.

The second stage of the research sought to investigate policy implementation, and focused on the local government level in Perth, Western Australia. Policy actors within planning, engineering, administrative and political roles at the local government level in Perth were interviewed about various barriers to policy implementation. The research illustrated a range of issues related to the implementation of policy in varying contexts of policy capacity. Local actors pointed to difficulties in how to implement what were considered to be vague State government policy directives. This was especially apparent in urban infill developments where multiple stakeholders were involved. Local government actors considered that State government policy was compromised by a lack of horizontal – or between-department – integration at the State level. Furthermore, the organizational structures of State government, characterized by separate departmental ‘silos’, increased the complexity and resource intensiveness of consultation and coordination of infrastructure provision.

This research demonstrates that the capacity of government to transition to new models of transport and mobility must be understood in terms of a wider concept of policy that incorporates how it is implemented in diverse contexts. Horizontal and vertical integration of policy across government is a necessary but not sufficient condition for successful uptake of new directions by all actors. It is critical in the process of developing such policy that statements of intent are transformed into implementable action, and that organizational structures are put in place to navigate the complexity and resource intensiveness of consultation and coordination processes.

Conclusion: what then is implied about making change happen?

The work discussed above suggests that there are serious complications in clearing a space for new pathways. There are multiple perspectives that give insight into the resilience of the current policy system. Today’s urban transport is the result of interwoven organizations and storylines that constitute the institutions of transport policy. Such interweaving is carried out not only in government policy and structures, but in the stories and technologies used by the community itself. The status quo is not maintained by a simple Machiavellian plot, where those who stand to gain impose their will. The logic of the current system is embedded in society. It is held fast by our understanding of risk, the structures of our organizations, the rationalizations we use to defend decisions and the technologies we privilege. Our conventional understanding of the system, and the technologies and rationalities we use to develop and maintain it presents us with a particular view of what transport is, one which it is difficult to see beyond. Even when

we manage to turn our gaze to a new vista and envisage a new system, the myriad of invisible lines pulling us back to the current path reorients us.

Creating a clearing for a new pathway for the transport system takes more than just a clear vision for the future. It will require careful understanding of the complexity of the weave holding the current system in place, and a willingness and ability to reweave the threads to maintain our focus on the new pathway. Perhaps in the final episode we will find ourselves guided mainly by those who can already knit the skeins differently through a different set of affordances. In any case, gaining permission for that future will require a merging of 'expert' and community knowledge through a well-designed consultation process. The shift to a new path is thus a complex task that will take the conscious efforts of many. It will have to be embedded vertically and horizontally not just in government policy and institutions but into the fabric of our cities and the stories that we tell ourselves about transport.

Notes

- 1 In this sense the Art of Government can be viewed as a particular type of Heideggerian episteme: a way of being which determines what we see and therefore what it is logical to do (Braun and Castree, 1998).
- 2 Perth's transport planning provides a possible exception where the demonstrated competencies of particular people allowed several rail construction projects to be implemented.
- 3 Actors bring tacit assumptions, beliefs and habits to the strategic games they engage in. The tacit sphere is the sphere of our thoughts and inter-subjective beliefs (Polanyi, M. 1976).